

Juan Monsalve

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Software Engineer specializing in applied AI and backend systems

Bucaramanga, Santander, Colombia

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I am a software engineer specializing in backend development, REST APIs, and applied AI systems. My experience includes designing scalable architectures and implementing containerized solutions using Docker and AWS. I recently built a local-first Spanish voice assistant, architecting a 4-stage intent router that optimizes LLM integration by keeping calls off the critical path. I have a proven ability to optimize databases and deploy high-performance microservices, including a FastAPI and PostgreSQL backend that achieved 99.9% uptime.

Jobs

Team Manager — Operations & Data Analysis

WW Funcrafters JWA LLC

November 2025 - August 2026

- Led a cross-functional team of 15+ employees, overseeing technical operations, quality assurance, and KPI achievement.
- Automated weekly performance reporting using Python and Pandas to analyze productivity metrics, increasing reporting efficiency by 40%.
- Managed escalated technical cases, performing root-cause analysis to deliver effective resolutions and maintain 95%+ operational target achievements.

Customer Technical Support Engineer

Concentrix Web Help

December 2023 - September 2024

- Diagnosed and resolved complex application errors and functional issues through systematic root-cause analysis and log debugging.
- Collaborated closely with L2/L3 engineering teams by documenting reproducible steps and technical logs for software bugs, accelerating time-to-resolution.
- Met quality and response-time KPIs consistently in a high-volume environment while adapting to continuous software updates and architectural changes.

Education

Git & GitHub

Self-studyCoursework

Python Programming

Self-studyCoursework

SQL & Databases

Self-studyCoursework

Linux Fundamentals
Self-studyCoursework

AWS Cloud Practitioner Fundamentals
Self-studyCoursework

Agile & Scrum
Self-studyCoursework

Systems Engineer (Bachelor's Degree)
Universidad Pontificia Bolivariana
January 2023 - January 2026

Projects

Goat Guard — Network Monitoring & Security Platform
University project
January 2026

- Developed a centralized network monitoring platform utilizing Python and system scripting to track hardware resources, bandwidth, and device availability across local networks.
- Designed real-time dashboards and implemented automated event detection mechanisms to trigger alerts for unauthorized network usage and performance bottlenecks.
- Architected the system for high scalability and proactive incident detection, improving local network security response times.

Apollo — Local-First Spanish Voice Assistant
Personal Project
January 2026

- Built a fully offline Spanish voice assistant that runs entirely on-device using Silero VAD, faster-whisper large-v3-turbo on CUDA, Piper TTS, and a local LLM served through Ollama.
- Designed a 4-stage intent router that resolves the majority of commands in ~2 ms and keeps the 250–500 ms LLM call off the critical path, measuring a 0.135 s end-to-end turn on the target hardware.
- Diagnosed and fixed a semantic-routing failure by mean-centering embeddings and introducing a negative fallback class, converting the decision from absolute to relative to improve performance.
- Constrained LLM execution behind a tool-call allowlist with strict Pydantic schemas to ensure the model only emits validated tool calls, supported by 307 unit tests.

NovaTrack — Scalable Backend API for Task Management
Personal Project
January 2026

- Designed and developed a high-performance RESTful API using Python (FastAPI) and PostgreSQL, implementing robust user authentication (JWT) and role-based access control.
 - Containerized the microservices architecture utilizing Docker and orchestrated seamless deployment on AWS EC2, achieving 99.9% application uptime.
 - Engineered automated CI/CD pipelines via GitHub Actions to streamline testing and deployment workflows, reducing code release cycles by 30%.
 - Integrated comprehensive API documentation using Swagger UI, facilitating efficient frontend integration.
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Languages

Spanish - Native or fully fluent

English - Conversational

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